

Interphase mass transfer Modelling 2025

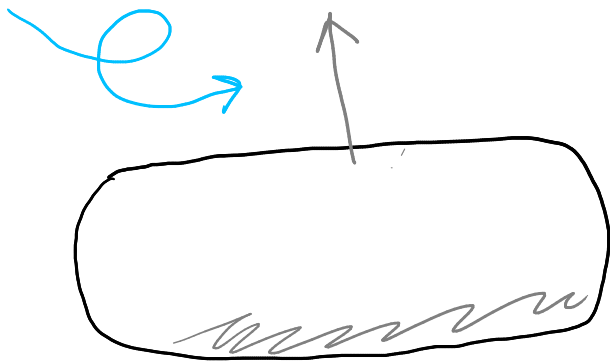
Some notes on interphase mass transfer

See also Cussler 2009: Section 8.5
That is an excellent introduction to interphase
mass transfer (or mass transfer through an
interface).

Interphase mass transfer - through an interface

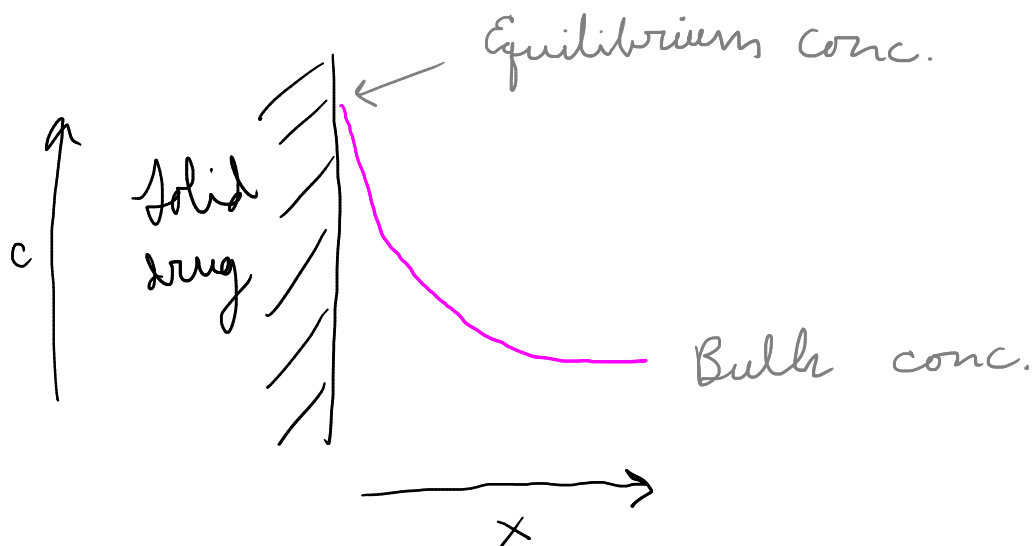
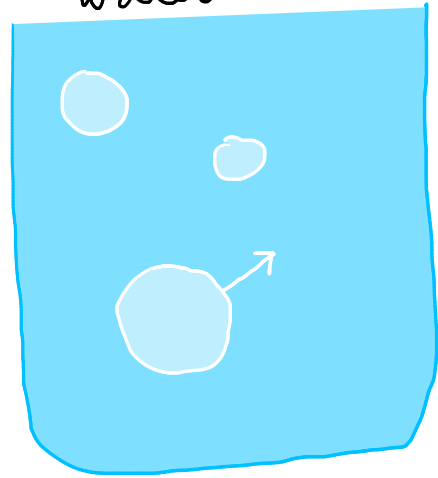
These two examples both have a pure phase!

Diffusion does not limit mass transfer in a pure phase



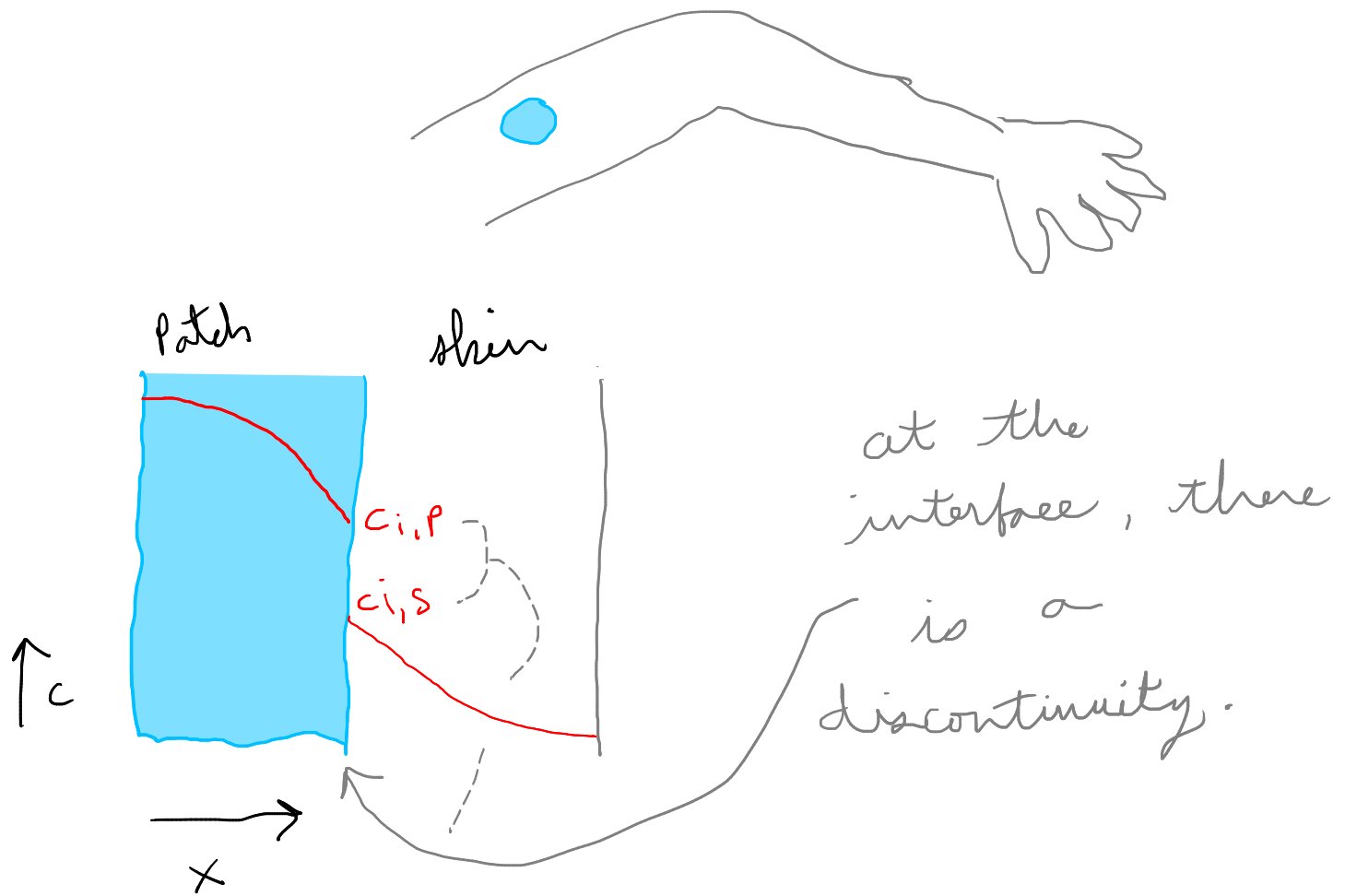
Drug dissolution into
water

O_2 or O_3 into
water



Mass transfer between two mixed phases is more common, and unfortunately more complicated

Interphase mass transfer

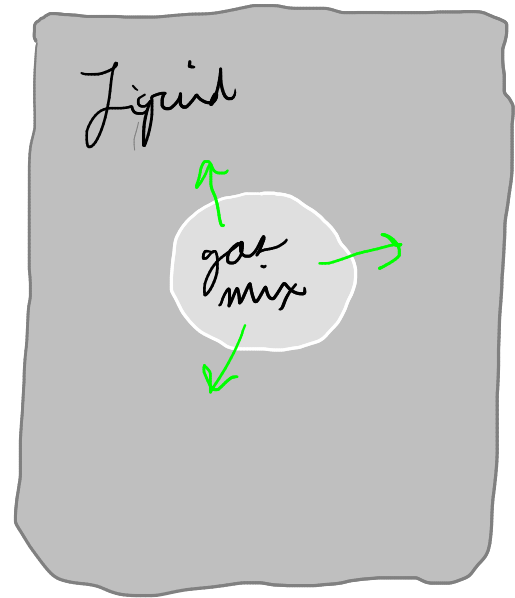
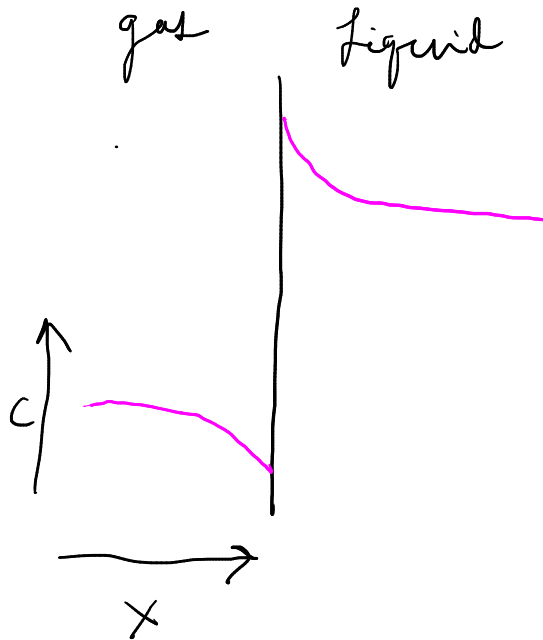


$$\frac{c_{i,p}}{c_{i,s}}$$

We assume equilibrium at the interface.

So the ratio is related to partition coefficients or solubility ratios.

another example, gas mixture
e.g. air with O_2



→
direction
of mass transfer

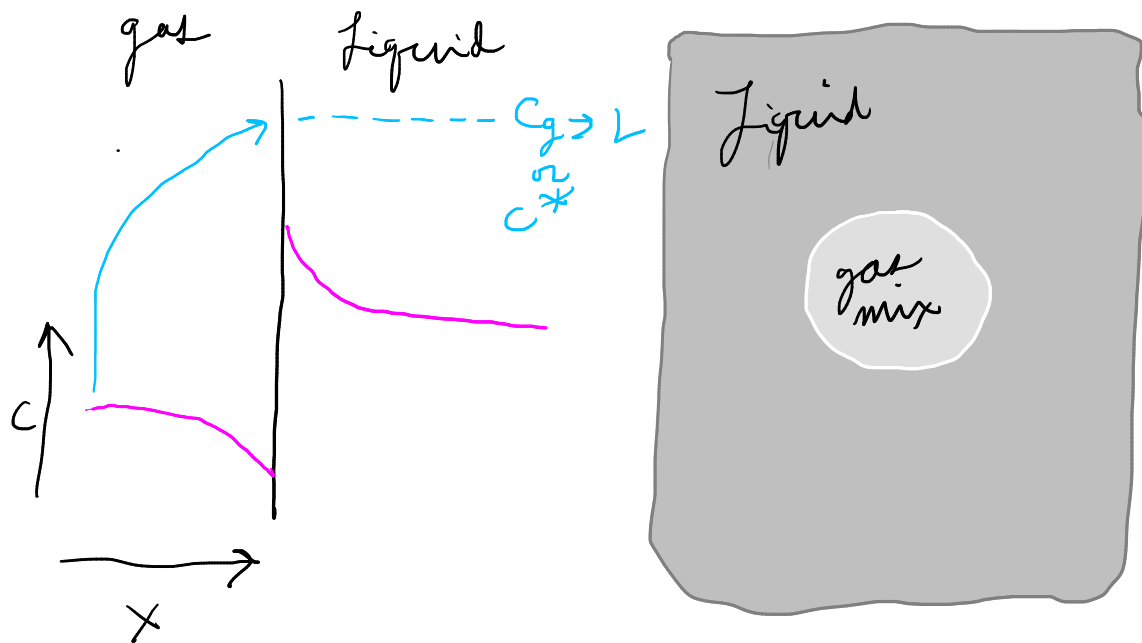
So how do we handle this complicated situation?

It is definitely NOT appropriate to use concentrations in different phases directly in a mass transfer equation

g_{O_2} / m^3_{air} cannot be directly compared to $g_{O_2} / m^3_{H_2O}$
to

Overall mass transfer coefficients

General solution is to express both concentrations in the units of a single phase



For oxygen, for example, it is common to use water phase concentration.

The actual gas phase concentration is converted into the concentration in water that would be in equilibrium with that gas phase concentration

see blue above - $C_{g \rightarrow L}$ or C^*
is the concentration that would be
in equilibrium with $CO_{2, gas}$.

So that gives us our gradient. Then the mass transfer coefficient can be calculated from some guess about film thickness, plus partitioning parameters (like Henry's law constant), and Fick's law (by someone else, not in this class!)

Or, it can be measured and correlated to e.g., fluid properties and diffusivity or mixing intensity etc. And we can find it in a table.

But, it is essential that the mass transfer coefficient is for the same phase used for concentrations!

So we have

Gas phase concentration converted to liquid phase units

$$J = K_L (c_{g \rightarrow l} - c_l)$$

$$= K_L (c^* - c_l)$$

/

Overall mass transfer coefficient in liquid phase units

Overall mass transfer coefficient in gas phase units

$$J = K_G (c_g - c_{l \rightarrow g})$$

Liquid phase concentration converted to gas phase units

Oxygen example

Overall mass transfer coeff.
in water phase units

$m \cdot s^{-1}$

$$J = k_L (C_{sat} - C_w) \quad (g \cdot m^{-2} \cdot s^{-1})$$

Gas concentration
or partial pressure
converted to water
phase concentration

Actual bulk concentration
in the water

$g_{O_2} m_{H_2O}^{-3}$

same units!

e.g. $P_{O_2} = 0.2 \text{ atm} \rightarrow 10 g_{O_2} m_{H_2O}^{-3}$

Conversion is done with
Henry's law constant for the
appropriate temperature

$k_L a$ includes bubble area available
for transport

$k_L a$ units: $m s^{-1} \cdot m^2 m^{-3} = s^{-1}$